

Grade 7
Unit 5
Lesson 6 Practice

Name _____
Date _____
Period _____

Shannon's TV, Figure D, has a height of 22 in. and width of 33 in. Shannon drew a scale model of her TV screen, Figure E, with a height of 4 in. width of 6 in. Use this information to complete problems 1-5.

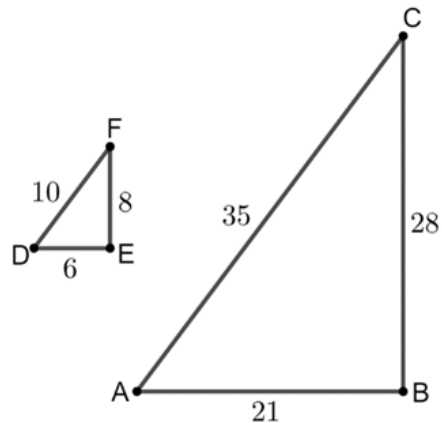


1. What is the scale factor from Figure D to Figure E?
2. What is the area of Shannon's TV screen?
3. What is the area of Shannon's scale model?
4. What is the ratio of the area of Figure E to the ratio of the area of Figure D?
5. What is the relationship between the scale factor and the ratios of the areas?

Right triangle DEF has a height of 8 units and base of 6 units. Right triangle ABC is a scaled-up image of right triangle DEF . Right Triangle ABC has a height of 28 units and base of 21 units. Use this information to complete questions 6-10.

6. Complete the table below.

	Triangle DEF	Triangle ABC
Perimeter		
Area		



7. What is the scale factor from Triangle DEF to Triangle ABC ?
8. How does the perimeter of Triangle DEF compare to Triangle ABC ?
9. How do the perimeters of Triangle DEF and Triangle ABC relate to the scale factor?
10. How do the ratios of the areas of Triangle DEF and Triangle ABC relate to the scale factor?